Posterior distributions

Outline

Topics

- Notion of posterior distribution.
- Examples
- How to use a posterior?

Rationale

The posterior distribution appears in the second step of the Bayesian Recipe and is therefore encountered (at least implicitly) in all full Bayesian problems.

Definition

The conditional PMF of the unknowns X given the observation Y is the called the **posterior PMF**.[1]

Notation

We use $\pi(x) = \mathbb{P}(X = x | Y = y)$ for the posterior PMF.

Recall:

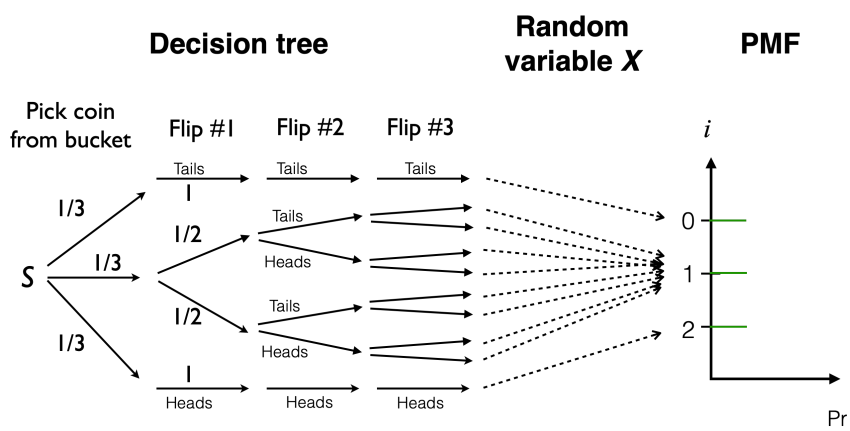
$$\pi(x) = \frac{\gamma(x)}{Z},$$

where $\gamma(x) = \mathbb{P}(X = x, Y = y)$ is the un-normalized posterior and $Z = \mathbb{P}(Y = y)$ is the normalization constant.

Examples

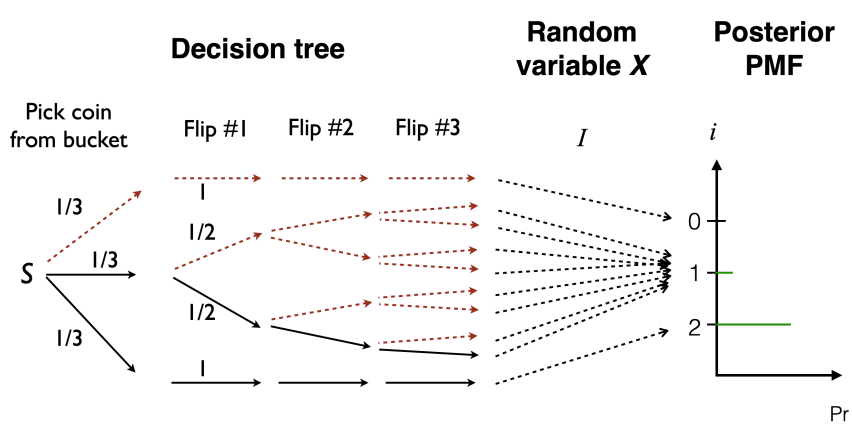
Example 1: coins in a bag

Visualization of the prior PMF and how it arises from the decision tree and random variable X (showing 3 flips instead of 4):



- Recall that the probability of each path is the product of the edge labels.

Visualization of the **posterior** PMF and how it arises from the decision tree and random variable X:



- Recall we zero out the contribution of the paths not compatible with the observation (Heads, Heads, Heads).
- This gives a list of numbers that do not sum to one, so we renormalize them.

Example 2: rocket insurance

You will construct prior and posterior PMFs in question 2 of this week's exercises.

What to do with a posterior?

- Show a visualiation (posterior PMF).
- Compute a **summary** of the PMF:
  - **point estimate:** single “best guess”, or
  - **a credible region:** a set of “guesses”.
- More generally: decision theory / Step 3 of the Bayesian Recipe.

Footnotes

1. if the unknown quantity is continuous, the posterior will be expressed using a density. A term that captures both the continuous and discrete case is “distribution” i.e. “posterior distribution”. ↩